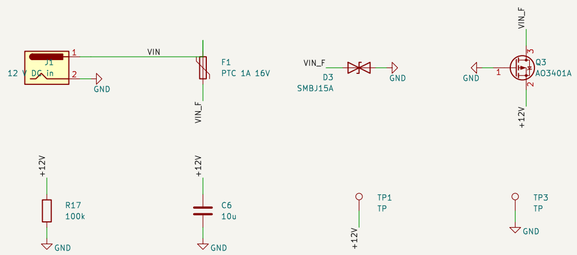
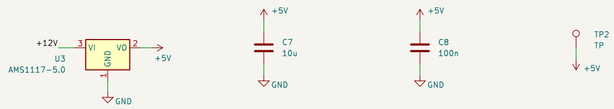
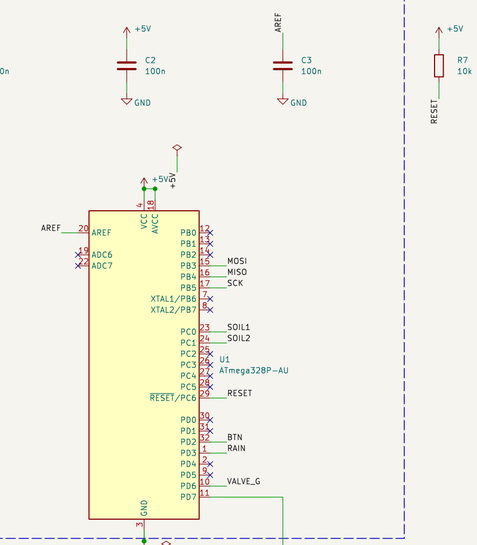
12 V INPUT: jack, PTC, TVS, reverse-polarity PMOS



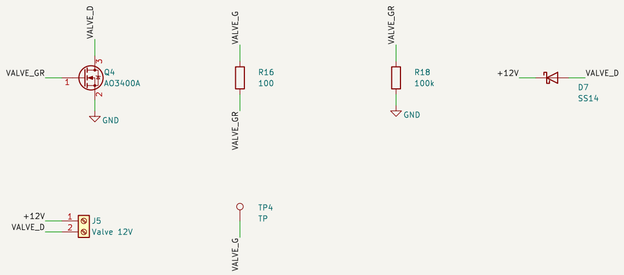
5 V REGULATOR



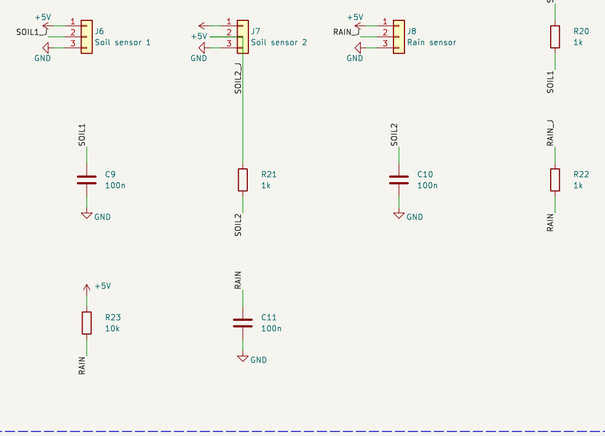
MCU: ATmega328P-AU



VALVE DRIVER: logic-level NMOS, flyback diode, terminal



SENSORS: 2 x soil (analog), rain (digital)



USER + PROGRAMMING



Valve: 12 V DC solenoid (about 0.3 A) switched low-side by Q4 with Schottky flyback D7. Q3 (AO3401A) blocks a reversed 12 V plug, D3 (SMBJ15A) clamps surges, F1 limits a fault to 1 A. AMS1117-5.0 drops 12 -> 5 V for about 40 mA of logic and sensors (about 0.3 W).

First engineering release; see pcb/README.md for status and limits

ClaudeWork

Sheet: /
File: kheti_mitra.kicad_sch

Title: Kheti-Mitra irrigation controller — main board

Size: A2	Date: 2026-11-01	**Rev: B**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1